

Deep Learning — Tutorial 3

Machine Learning Basics (Chapter 5)

Practice set — not for submission

1 Learning Algorithms and Linear Regression

Q1. A learning algorithm is specified by a task T , a performance measure P , and an experience E . For an email spam filter, name a suitable T , P , and E . Is this supervised or unsupervised learning, and why?

Q2. Linear regression predicts $\hat{y} = \mathbf{w}^\top \mathbf{x}$. With design matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ (one example per row) and targets $\mathbf{y} \in \mathbb{R}^m$, the training error is $\text{MSE}_{\text{train}} = \frac{1}{m} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$. Derive, from $\nabla_{\mathbf{w}} \text{MSE}_{\text{train}} = \mathbf{0}$, the normal equations for the optimal $\mathbf{w}$.

Q3. Fit $\hat{y} = w_0 + w_1x$ to the data $(x, y) = (1, 2), (2, 2), (3, 4)$ using the normal equations. Give $\mathbf{w}$ and the fitted line.

2 Capacity, Overfitting and Underfitting

Q4. Define *generalization* (test) error and the *generalization gap*. Explain, in terms of model capacity, what distinguishes *underfitting* from *overfitting*.

Q5. A model achieves 1% training error but 12% test error. Which regime is it in? Give two concrete changes that would be expected to reduce the test error, and say which quantity each one targets.

Q6. You fit polynomials of degree d to a data set. Explain how d controls capacity, and state which of $d = 1$, $d = 4$, $d = 20$ (for, say, 15 noisy points from a mild curve) is most likely to underfit and which to overfit.

3 Regularization (Weight Decay)

Q7. Weight decay minimises $J(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \mathbf{w}^\top \mathbf{w}$ with $\lambda \geq 0$. Derive the modified normal equations, and hence the closed-form solution for $\mathbf{w}$.

Q8. Using your result from the previous question, describe what happens to the solution $\mathbf{w}$ in the two limits $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. What role does λ play as a hyperparameter, and why must it be chosen on a validation set rather than the training set?

4 Estimators: Bias, Variance and Standard Error

Q9. Let $x_1, \dots, x_m$ be i.i.d. with mean μ . Show that the sample mean $\hat{\mu} = \frac{1}{m} \sum_i x_i$ is an *unbiased* estimator of μ .

Q10. For the same data with variance σ^2 , the maximum-likelihood variance estimator is $\hat{\sigma}^2 = \frac{1}{m} \sum_i (x_i - \hat{\mu})^2$. It is known that $\mathbb{E}[\hat{\sigma}^2] = \frac{m-1}{m} \sigma^2$. (a) Is $\hat{\sigma}^2$ biased? In which direction? (b) Write the standard *unbiased* estimator. (c) By what factor does $\hat{\sigma}^2$ under/over-estimate σ^2 when $m = 5$?

Q11. The standard error of the mean is $\text{SE}(\hat{\mu}) = \sigma/\sqrt{m}$. If $\sigma = 10$ and $m = 100$, compute $\text{SE}(\hat{\mu})$ and give an approximate 95% confidence interval half-width for the mean.

Q12. For any estimator $\hat{\theta}$ of a fixed quantity θ , prove the decomposition

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}).$$

5 Maximum Likelihood Estimation

Q13. Data $x_1, \dots, x_m$ are modelled as i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Write the log-likelihood and derive the maximum-likelihood estimator $\hat{\mu}$.

Q14. A coin is flipped $n = 10$ times, landing heads $k = 7$ times, modelled as i.i.d. Bernoulli(θ). Write the log-likelihood and derive the MLE $\hat{\theta}$; give its numerical value.

Q15. Model the targets as $p(y | \mathbf{x}) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$. Show that maximising the conditional log-likelihood $\sum_i \log p(y_i | \mathbf{x}_i)$ over $\mathbf{w}$ is equivalent to minimising the mean squared error.

6 Bayesian Statistics and MAP Estimation

Q16. Place a Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ on the weights of the regression model $p(y | \mathbf{x}) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$. Show that the MAP estimate $\hat{\mathbf{w}}_{\text{MAP}} = \arg \max_{\mathbf{w}} \log p(\mathbf{w} | \text{data})$ is exactly a weight-decay (ridge) solution, and identify λ in terms of σ^2 and τ^2 .

Q17. In one or two sentences each, contrast the frequentist maximum-likelihood approach with the Bayesian approach: what does each treat as fixed vs. random, and what does the Bayesian method produce that a single ML point estimate does not?

7 Principal Components Analysis

Q18. Consider the four points $(1, 2)$, $(3, 3)$, $(5, 4)$, $(7, 5)$. (a) Compute the mean and centre the data. (b) Form the 2×2 covariance matrix (use the $\frac{1}{m}$ convention). (c) Find its eigenvalues and the first principal component (unit vector). (d) One eigenvalue is 0. What does this tell you about the intrinsic dimensionality of the data?

Q19. State the two equivalent objectives PCA optimises (in terms of variance and of reconstruction error), and say which matrix's eigenvectors give the principal components.

8 Stochastic Gradient Descent

Q20. Write the stochastic gradient descent update for a cost that is a mean over m training examples, using a minibatch of size $m' \ll m$. Explain why the cost of one SGD step is essentially independent of the full training-set size m .

Q21. Consider the model $\hat{y} = w x$ (scalar w) with per-example loss $L = (wx - y)^2$. Starting from $w = 0$ with learning rate $\varepsilon = 0.1$, perform one SGD step on the minibatch $\{(x=1, y=2), (x=2, y=2)\}$. Give the minibatch gradient and the updated w .

9 Conceptual Questions — the Big Picture

These questions have little or no arithmetic. Answer them in a few sentences each; together they capture the essence of the chapter and why it motivates deep learning.

Q22. State the *No Free Lunch* theorem informally, and explain its practical implication: if no algorithm is universally best, what must a good machine-learning method do instead?

Q23. Explain the *curse of dimensionality*: why does the number of distinct configurations of a space grow so fast with dimension, and why does this make purely local methods (e.g. nearest neighbours) struggle in high dimensions?

Q24. Many classical methods rely on a *local constancy* / smoothness prior — the assumption that the target function changes little within a small neighbourhood. Explain why this prior alone is insufficient to learn complicated functions in high dimensions.

Q25. State the *manifold hypothesis*. Why does the assumption that data concentrates near a low-dimensional manifold make otherwise-hard high-dimensional learning problems tractable?

Q26. Tie the chapter together: explain how *capacity*, *regularization*, *bias–variance*, and the choice between *maximum likelihood* and *MAP* are all really different views of the same central goal — controlling *generalization*.